

Week 9 Tutorial MEMO

Question 1

a) Upward trend, seasonality and random components.

b)

i) $\hat{y}_t = 68.75 + 0.5809t$

ii)

t	y_t	CMA(4)t	$\frac{y_t}{\hat{y}_t}$
1	85		
2	54		
3	57	68.5	0.832
4	75		
5	91		
6	62		
7	61	73.135	0.834
8	78		
9	92		
10	62		
11	67	76.75	0.873
12	83		
13	98		
14	67		
15	68		
16	79		
		Average	0.846

iii) On average the first quarter employment figures are 25.6% above the annual average.

c) $\hat{y}_{19} = 68.75 + 0.5809(19) = R79.77$ million (only trend)

$\hat{y}_{19} = [68.75 + 0.5809(19)]0.846 = R67.50$ million (with seasonality)

Question 2

a) Seasonality and slight trend (random component)

b) Values are as follows:

- A. 87.125
- B. 0.6083
- C. 0.5802 (or 0.5801 due to rounding)
- D. 91.3526 (or 91.3478 due to rounding)
- E. 1.3961
- F. 1.3990
- G. 0.5817

c) $\hat{y} = (77.176 + 2.129 \times 19) \times 0.9892 = 117.627 \times 0.9892 = 116.3566$